

79,979

80,000

116,744

141,000+

5 DEATHS

In Every Single Hour in India

70% of cases are diagnosed at advanced Stage III or IV — when survival
crashes below 20%.

2.1 MILLION+

People Died Due to Oral Cancer in the Past Decade



#1 GLOBAL ORAL CANCER BURDEN

India Is the Largest Country with the Highest Number of Oral Cancer Cases

India accounts for over one-third of the global oral cancer incidence and mortality, making it the highest in the world.

#1 WORLDWIDE

Highest global oral cancer mortality burden.

250 MILLION+

Active chewable tobacco and areca nut consumers.

Team *BraKet Reloaded*

Pioneering Hybrid Quantum AI for Preventive Oral Oncology in India

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What Is *Oral Cancer*?

Oral cancer is a malignant growth that develops in the tissues of the oral cavity and oropharynx, predominantly as Oral Squamous Cell Carcinoma (OSCC).

What Is *Oral Cancer*?

ANATOMICAL SITES

Mucosal Lining

Arises primarily on the lateral tongue, buccal mucosa (inner cheeks), floor of the mouth, gums, and hard palate.

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Silent Lesions

Begins as subtle white patches (Leukoplakia), red patches (Erythroplakia), or stiffened tissue (Oral Submucous Fibrosis).

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MALIGNANT PROGRESSION

Deep Tissue Invasion

Without early intervention, dysplasia penetrates the basement membrane into deep muscle and spreads rapidly to lymph nodes.

The Major *Causes*

Over 90% of oral cancers in India are directly attributable to chewable and smoked carcinogens.

The Major *Causes*

PRIMARY DRIVER

Tobacco & Khaini

Chewing tobacco delivers high concentrations of carcinogenic Tobacco-Specific Nitrosamines (TSNAs), causing severe DNA damage.

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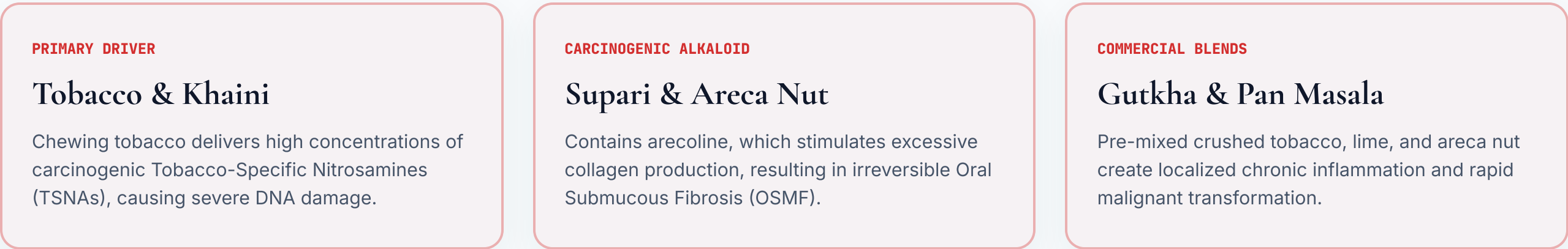
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CARCINOGENIC ALKALOID

Supari & Areca Nut

Contains arecoline, which stimulates excessive collagen production, resulting in irreversible Oral Submucous Fibrosis (OSMF).

The Major *Causes*



The *6-Month* Warning Window

Oral cancer gives the human body a 6-month warning window in plain sight.
Yet in India, **70% of patients waste this window** because no one is looking
and there is no accessible tool to triage it.

DAY 0 - 90 • EARLY WINDOW

Tiny Painless Patch

Visible white patch (Leukoplakia) or localized tissue stiffness.
Asymptomatic, non-ulcerated mucosal dysplasia.

100% CURABLE with ~₹1,000 outpatient care

DAY 180 • LATE PROGRESSION

Deep Muscle Invasion

Dysplasia penetrates basement membrane into deep jaw tissue.
Lymph node metastasis and bone erosion.

SURVIVAL < 20% with radical jaw resection

The Two Futures: *A Stark Contrast*

The heartbreaking difference between screening today versus discovering it 6 months later.

STAGE 0 / PRECANCER

Early Detection

CLINICAL TREATMENT

15-minute topical outpatient care / habit cessation

SURVIVAL RATE

> **90% SURVIVAL**

FINANCIAL COST

₹199 – ₹1,500

QUALITY OF LIFE

100% normal speech, eating & smile preserved

STAGE III / IV METASTASIS

Late-Stage Discovery

CLINICAL TREATMENT

Radical jawbone resection (cutting half the face) + chemo

SURVIVAL RATE

< **20% SURVIVAL**

FINANCIAL COST

₹8,00,000 – ₹15,00,000 (Bankrupts families)

QUALITY OF LIFE

Severe facial disfigurement & lifelong feeding tubes

Is There Any Cure for Late-Stage Oral Cancer?

UNFORTUNATELY, NO.

*Once it advances to Stage III or IV,
survival collapses below 20%.*

Late-stage treatments require radical surgery, jaw bone resection,
and facial disfigurement, with devastating physical, psychological,
and financial tolls on families.

EARLY DETECTION IS THE ONLY SOLUTION.

Survival Exceeds 85% When Detected at Stage I or Precancer.

Simple, non-invasive outpatient treatment cures precancerous lesions completely without surgery or disfigurement — if identified in time.

JUST 900

Functional Cancer Care Facilities in All of India

Specialized oncology centers equipped with radiation bunkers,
surgical oncology suites, and dedicated oral pathology labs.

If We Do a Little *Math...*

$$\begin{array}{ccc} 147.8 \text{ Crore} & & 900 \\ \text{Population} & \div & \text{Facilities} \\ & & \\ & = & 1,642,222 \end{array}$$

Over 1.64 Million Citizens per Single Cancer Center

How can 900 centers screen and triage 147.8 crore citizens? *It is physically impossible without decentralized, smartphone-based AI triage at point-of-care.*

For Getting an Early Detection Test...

You Need to Wait in a Queue of

1,642,222

Patients Per Cancer Facility

By the time a patient's turn arrives in this queue, an easily curable precancerous lesion has already advanced into late-stage metastatic carcinoma.

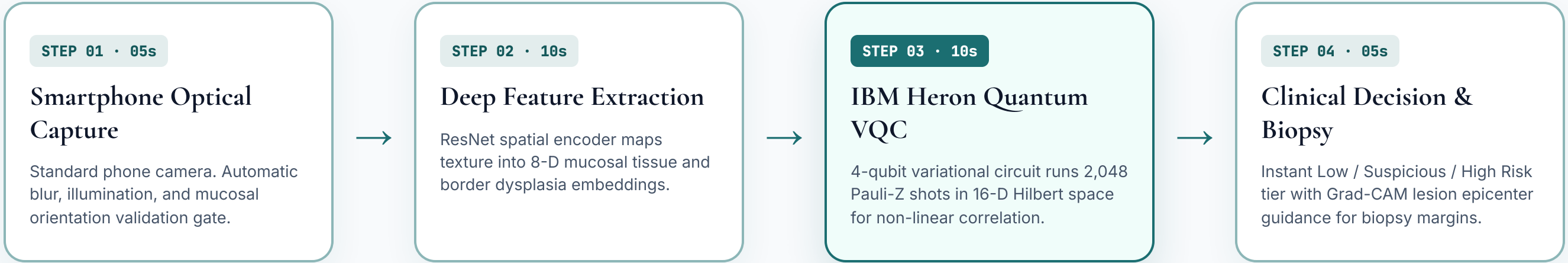
Is There *Any* Solution?





*India's First Hybrid QML Oral
Cancer Detection Platform*

Point-of-Care Triage in *30 Seconds*



Live Platform *Demonstration*



**What is the Cost of Per
Scan?**

RS 4,999

~~Rs 4,999~~

JUST Rs 199

The "*Doctor Distance*" Paradox

Why ₹199 smartphone triage changes everything for rural India.

THE GEOGRAPHIC DIVIDE

70% Rural vs 95% Metro

70% of India's population resides in rural and semi-urban areas, yet **95% of qualified oncologists and cancer hospitals** are concentrated in 8 Tier-1 metro cities.

THE DAILY WAGE REALITY

14-Hour Train to Metro

A daily wage worker in rural Bihar, UP, or Assam cannot afford to lose 3 days of wages and travel 14 hours for a hospital checkup. **QuOra brings the oncologist to their local clinic for ₹199.**

Economic Drain on *India's* *Working Class*

The macro-economic crisis caused by late-stage oral cancer treatment.

DEMOGRAPHIC IMPACT

Age 30 to 55

Oral cancer in India primarily strikes people in their **prime working years**—the sole breadwinners supporting young children and elderly parents.

CATASTROPHIC DEBT

80% Out-of-Pocket

Over 80% of cancer treatments in India are paid out-of-pocket, pushing **60 Million Indians below the poverty line** every single year.

THE ECONOMIC ROI

₹199 vs ₹10 Lakhs

A **₹199 optical scan** prevents a **₹10 Lakh catastrophic financial wipeout**, saving both human lives and household livelihoods.

For Hospitals & Clinics: *The QuOra Webapp*

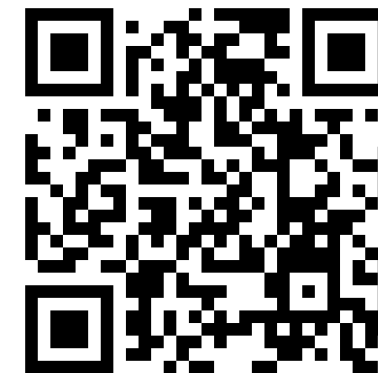
For hospitals, diagnostic labs, and dental surgery centers, we deliver clinical-grade decision support with real-time triage queue and Grad-CAM spatial overlays.

VISIT DIRECT ONLINE

imperialocietyofsciences.space

OR SCAN THE ATTACHED QR

Instant workstation & clinical browser launch.



imperialocietyofsciences.space

And for Normal Citizens: *The QuOra App*

Empowering every citizen with pocket-sized screening, personalized habit reduction coaching, and one-touch clinic navigation.

30-SECOND OPTICAL SELF-SCREEN

Instant AI risk assessment and guidance in local Indian languages.

HABIT CESSATION TRACKER

Daily monitoring to halt and reverse chronic submucous fibrosis.



Scan to Download App

How to Use *The QuOra App*

Designed for zero medical training — point, align, scan, and receive clinical guidance.

STEP 01

Open Viewfinder

Launch the app and point your smartphone camera toward the oral cavity under good natural or room lighting.

STEP 02

Smart Alignment

Real-time mucosal bounding box automatically aligns to tongue, inner cheeks, gums, or suspicious white/red lesions.

STEP 03

30s Quantum Scan

Neural backbone and IBM Heron quantum variational circuit analyze tissue erythema, border stiffness, and dysplasia.

STEP 04

Actionable Report

Instant Green / Amber / Red triage score with 1-touch connection to nearest accredited dental oncologist.

Our *Business Model*

High-margin software revenue across institutional B2B and consumer B2C channels.

B2B INSTITUTIONAL Web Subscription ₹15,000 – ₹45,000 / MONTH Per clinic & diagnostic laboratory. Unlimited optical scans, Grad-CAM overlays, and direct PACS / EHR integration.	B2C CONSUMER App Scans (Freemium) ₹99 / REPORT OR ₹499 / YR First scan 100% FREE. ₹99 per comprehensive AI risk assessment or ₹499 annual pass for unlimited family monitoring.	CARE NAVIGATION Tele-Consultation 15% – 20% COMMISSION Platform referral fee on high-risk oncology tele-consultations and expedited biopsy appointments booked in-app.
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Hybrid *Quantum AI* Technology

QuOra is a dual-platform early oral cancer screening solution fusing classical vision with superconducting quantum circuits.

CLASSICAL BACKBONE

ResNet-18 Spatial CNN

Extracts rich 8-dimensional mucosal texture embeddings, boundary gradients, and localized tissue erythema patterns from optical photos.

QUANTUM PROCESSOR

IBM Heron r2 VQC

16-qubit Variational Quantum Circuit executes 2,048 Pauli-Z shots in Hilbert space to capture complex non-linear dysplasia correlations classical models miss.

Mobile app handles consumer pre-screening, while the web portal delivers clinical-grade diagnostic reports to labs and pathologists.

Target Pain Points & *Fundamental Solution*

CRITICAL PAIN POINTS

The Diagnostic Vacuum

1.64M citizens per cancer center with zero rural specialists. Early lesions (Leukoplakia/Erythroplakia) are mistaken for benign ulcers until late Stage III/IV when survival crashes under 20%.

FUNDAMENTAL SOLUTION

Zero-Hardware Point-of-Care Triage

Shifts diagnosis from late-stage emergency to curable Stage 0/I precancer. Runs on standard consumer smartphones + IBM Heron QPU cloud, eliminating the 1.6M queue bottleneck.

Market Size & *Growth Dynamics*

India represents the world's highest density growth opportunity for point-of-care AI oncology diagnostics.

INDIA GROWTH VELOCITY

22.5% CAGR

Indian AI Healthcare & Point-of-Care Diagnostics market expansion, driven by Ayushman Bharat digital health missions and nationwide tele-screening initiatives.

GLOBAL AI DIAGNOSTICS SECTOR

14.2% CAGR

Global AI/software oncology screening tools growth (overall global medical diagnostics expanding at 6.8% CAGR).

Total Addressable Market (*TAM*)

₹40,000 CR

Global Oral Cancer Diagnostics & Screening Market

Worldwide diagnostic expenditure across point-of-care screening,
histopathology validation, and recurring oncology monitoring.

Serviceable Addressable Market (SAM)

₹10,000 CR

India & South-East Asia High-Risk Cohort

Directly targeting ~143,000 annual incident cases in India and 250 Million+ active chewable tobacco, khaini, and supari consumers.

Serviceable Obtainable Market (SOM)

₹150 CR

1.5% Market Capture in India (Years 1–3)

Achieved through onboarding 15,000 Primary Health Centres (PHCs), CSR-backed mass screening drives, and 45,000 dental clinics.

Customer Acquisition (*GTM*)

B2B: PATHOLOGISTS & LABS

Diagnostic Chain Integration

Direct integration with major diagnostic networks (**Dr. Lal PathLabs, SRL Diagnostics**) and dental clinics via free API trial tiers and clinical research partnerships.

B2C: HIGH-RISK CONSUMERS

Community Screening & CSR Drives

Targeted digital awareness for tobacco and betel nut users, paired with CSR-sponsored community screening camps featuring instant QR code app downloads.

ZERO DEATHS

*With the Completion of the
National Quantum Mission by
2030...*

We want to achieve ZERO preventable deaths caused by oral cancer in India.